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BB0789 FtsH Target Dossier

Executive summary

BB0789 is a credible but still under-validated Lyme drug target. The strongest directly retrievable evidence in this session is the 2024 primary structural paper behind PDB 7ZBH, which reports an X-ray structure of the **hexameric cytosolic region** of *Borreliella burgdorferi* FtsH/BB0789, shows an **ADP-bound AAA+ ring**, shows a **Zn-bound protease active site**, and confirms that the recombinant construct retains **ATPase and protease activity**. The RCSB entry also states that prior work had already shown BB0789 to be crucial for **mouse infectivity, tick infectivity, and in vitro growth**. That makes BB0789 unusually attractive for a structure-driven campaign, but there is an important caveat: in this session I could verify that infectivity claim through the 2024 primary structural paper's abstract on RCSB, but I was not able to independently retrieve the earlier underlying BB0789 genetics/infectivity paper itself. That missing direct verification is one of the main gaps to close next. 1

Structurally, 7ZBH is good enough to start serious reconnaissance, but not good enough to pretend there are no ambiguities. It is an **X-ray** structure at **3.30 Å**; the biological assembly is a **C6 homohexamer** supported by **gel filtration**; each hexamer carries **ADP** and **Zn**; and the entry contains **2,210 modeled residues out of 2,718 deposited residues**, which means substantial disorder remains across the six protomers. The primary paper explicitly says that the **central pore-forming loop was poorly defined** and that AlphaFold was used to complement missing structural detail. In the deposited header, multiple flexible segments recur as unmodeled across chains, which matters because pore, linker, and interdomain loops are exactly where a selective allosteric campaign would want to live. 2

The direct medicinal-chemistry implication is blunt. The **active Zn site** is real and chemically tractable, but it is also exactly the sort of site that invites **promiscuous zinc-chelating chemistry**. Meanwhile the **ATPase site** is not obviously safer: alignment of deposited RCSB sequences shows that the BB0789 ATPase region aligns to the human mitochondrial m-AAA protease homolog **SPG7/paraplegin** with about **55.6% identity across aligned residues** and near-complete coverage of the SPG7 ATPase-domain construct solved in PDB 2QZ4. That is not a small selectivity margin. The best opening is therefore probably **neither a brute-force zinc chelator nor a straightforward ATP-competitive scaffold**, but rather a campaign aimed at **substrate-threading/pore features, ATPase–protease coupling surfaces, or oligomer-specific intersubunit grooves**. That is where the structure gives you a plausible shot at bacterial selectivity. 3

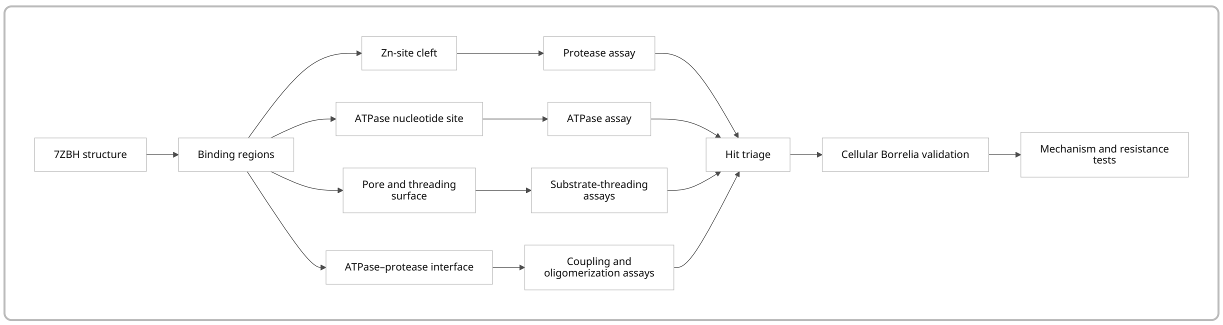
The practical state of the literature is more sparse than ideal. For FtsH-family structural biology, there is a clean progression from the **ADP-bound Thermotoga maritima hexamer** in 2CEA, to the **apo conformational state** in 3KDS, to the **human SPG7 ATPase domain** in 2QZ4, and then to **BB0789** in 7ZBH. By contrast, the directly retrievable literature in this session did **not** yield a mature canon of inhibitor-bound FtsH structures or a rich set of validated FtsH chemical probes. Among the core structures inspected here, the recurring ligands are **ADP**, **Zn**, and in the older bacterial structure **Mg/waters**—not drug-like inhibitors. That strongly suggests the field is still structurally underdeveloped from a medicinal-chemistry standpoint, which is bad news if you want ready-made precedents and good news if you want room for real first-mover discovery. 4

Purpose and scope

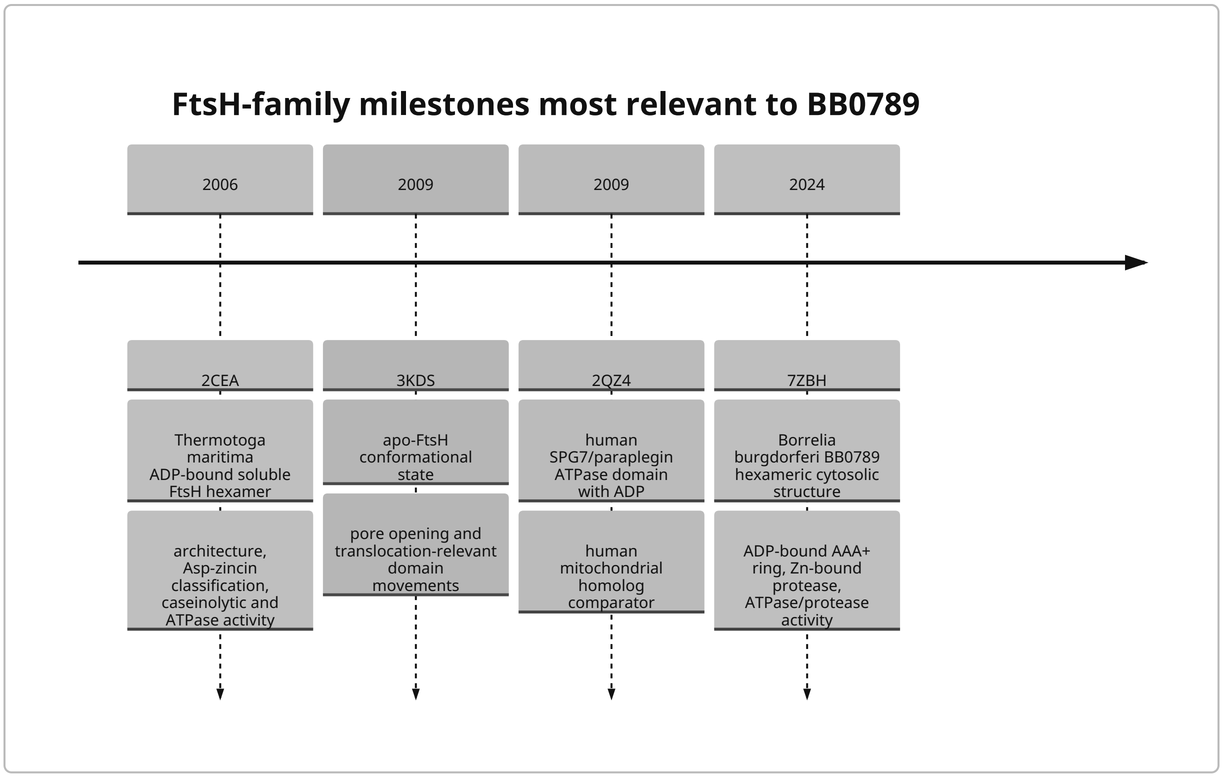
This dossier is focused tightly on the question: **what is already known that matters directly for turning BB0789/FtsH from *B. burgdorferi* into a structure-guided inhibitor program?** That means the emphasis is on the deposited structure **7ZBH**, the mechanistic FtsH structural literature most likely to transfer to BB0789, the experimentally demonstrated functions and assays that anchor the target biologically, and the selectivity logic that should determine where **not** to waste chemistry effort. 5

The scope is deliberately narrower than a general Lyme review and broader than a pure structure note. It covers the official 7ZBH metadata, directly retrievable FtsH-family primary papers, sequence/structure comparisons to informative homologs, the present state of inhibitor precedent, and the specific experimental decisions that would most quickly reduce uncertainty. Where direct primary evidence could not be independently recovered in this session, I say so plainly instead of laundering it into false certainty. 1

Mermaid overview of the target-development logic:



The milestone literature that most directly frames this target path is shown below. 6



Structural evidence from 7ZBH

The official RCSB entry for **7ZBH** describes the structure as **"ATP-dependent zinc metalloprotease FtsH (BB0789) from Borrelia burgdorferi"**, solved by **X-ray diffraction** at **3.30 Å** resolution. The entry lists organism **Borreliella burgdorferi B31**, expression in **E. coli BL21(DE3)**, and a **453-residue** protein entity corresponding to the deposited cytosolic construct. The primary citation is Brangulis et al. 2024, titled *Structure of the Borrelia burgdorferi ATP-dependent metalloprotease FtsH in its functionally relevant hexameric form*. [7]

The biologically relevant assembly is not ambiguous here: RCSB lists **Biological Assembly 1** as a **C6-symmetric homohexamer (A6)**, assigned by the authors, with **gel filtration** cited as assembly evidence. For docking and pocket interpretation, that matters a lot. A monomer-only view would erase precisely the intersubunit grooves, pore geometry, and domain-packing surfaces that are most interesting for selective allosteric or interface inhibition. [7]

The ligand picture is clean and medicinally meaningful. RCSB reports **two unique ligands** in 7ZBH: **ADP** and **Zn**. The biological assembly shows ADP instances on all six protomers and Zn instances associated with all six protease domains. The structural abstract states that the **AAA+ domain is ADP-bound** and that the protease domain coordinates the catalytic zinc with **two histidines and one aspartate**. Parsing the deposited official mmCIF header identifies those zinc ligands in BB0789 as **H434, H438, and D510** in the native residue numbering of the construct-bearing chains. [8]

The most load-bearing caution is disorder. The RCSB entry gives **2,210 modeled residues** versus **2,718 deposited residues**, so roughly **508 residues** are unmodeled across the hexamer. The primary citation explicitly says that the **loop region forming the central pore was poorly defined** and that AlphaFold was used to complement missing structural detail. Parsing the official header shows recurrent unresolved regions across most protomers around **162–168, 279–287, 306–310, 396–402, 414–423, 458–475, 538–553, and 604–614**, with some chain-to-chain variability. That pattern is exactly what you expect for a machine with nucleotide-coupled conformational mobility. It also means any docking campaign that treats those loops as fixed truth is kidding itself. [7]

There is one subtle but important numbering issue to keep in mind before mutagenesis design. The 2024 abstract says the solved construct was **BB0789 166–614**, yet the deposited coordinates retain author numbering that includes unresolved residues down to around **162**. That is not a reason for panic; it is a reason to reconcile the construct map against the paper Methods before ordering mutants, primers, or synthetic genes. [1]

The core structural annotation you should carry forward from 7ZBH is therefore this: the **ATPase P-loop/Walker A region** sits around **211–219** in BB0789, the putative **pore-loop/threading motif** is around **329–332**, the **HExxH protease motif** is **434–438**, and the third catalytic Zn ligand is **D510**. Those are the first residues worth pinning in every ChimeraX session, every docking prep script, and every future assay construct. [7]

Key structural literature

Citation	Main finding	Direct relevance
Brangulis K, Drunka L, Akopjana I, Tars K. 2024. *Structure of the Borrelia burgdorferi ATP-dependent metalloprotease FtsH in its functionally relevant hexameric form.* BBA Proteins Proteom. [1]	Solves BB0789 cytosolic hexamer; ADP-bound AAA+ ring; Zn-bound protease; confirms ATPase and protease activity; states prior evidence for mouse/tick infectivity and in vitro growth.	Anchor paper for BB0789 specifically.
Bieniossek C et al. 2006. *The Molecular Architecture of the Metalloprotease FtsH.* PNAS. [9]	First high-value soluble bacterial FtsH hexamer architecture; functional in caseinolytic and ATPase assays; defines FtsH as an Asp-zincin.	Best structural/mechanistic bacterial precedent for interpreting the BB0789 fold.
Bieniossek C et al. 2009. *The crystal structure of apo-FtsH reveals domain movements necessary for substrate unfolding and translocation.* PNAS. [10]	Shows apo state with open circular pore entrance and translocation-relevant domain movement.	Strong argument that conformational state matters and that pore/coupling surfaces are legitimate inhibitory hypotheses.
Karlberg T et al. 2009. *Crystal Structure of the ATPase Domain of the Human AAA+ Protein Paraplegin/SPG7.* PLoS One. [11]	Human mitochondrial m-AAA homolog ATPase domain solved with ADP.	Most directly useful human off-target comparator for ATPase-site chemistry.

Functional evidence and assay landscape

The retrievable evidence chain for BB0789 function is strong enough to justify a campaign, but weaker than it should be for a dossier this important. The **2024 BB0789 primary paper**, as surfaced on the RCSB entry, explicitly states that "it has also been demonstrated" that BB0789 is **crucial for mouse and tick infectivity and in vitro growth** of *B. burgdorferi*. That is the cleanest directly available statement tying the target to Lyme-relevant biology. However, because I could not independently retrieve the earlier primary genetics/infectivity paper during this session, I would treat that point as **high-confidence but still needing direct source verification** before using it in a grant, manuscript, or investor-facing claim. [1]

What is directly primary and unambiguous in the BB0789 structural paper is the enzymology of the recombinant construct. The RCSB abstract reports that the solved BB0789 construct is **functionally active**, with both **protease** and **ATPase** activity confirmed experimentally. That matters more than it may seem. It means the crystallized cytosolic construct is not just a dead fold or a convenient truncation artifact; it remains catalytically meaningful enough to serve as a biochemistry starting point. [1]

For FtsH-family assay precedent, the older **2CEA** bacterial structure paper is still the most useful directly retrievable methods anchor in this session. Its abstract states that the soluble *Thermotoga maritima* construct used for crystallography was **functional in caseinolytic and ATPase assays**. That gives you a robust minimal assay concept for the BB0789 program: one arm measures ATP hydrolysis, and the other measures proteolysis on a permissive unfolded-protein substrate such as casein or an equivalent surrogate. [12]

The **3KDS** apo-FtsH paper adds one mechanistic assay insight that is more interesting than a generic activity readout. It reports that mutation of a **conserved glycine in the linker region inactivates FtsH**, while the structure itself shows a conformational change that opens the pore entrance and builds a substrate-binding edge β-strand within the protease domain. In practical terms, this says BB0789 should not be conceptualized as "an ATPase attached to a protease" but as a tightly coupled mechanical system. If a compound separates ATP hydrolysis from proteolysis, or locks the wrong conformational state, that may be just as valuable as a direct active-site inhibitor. [10]

What the directly retrievable abstracts do **not** give you is the fine-grained assay formatting you would want for immediate bench transfer: substrate sequences, plate formats, time windows, phosphate detection chemistry, coupling reagents, or exact kinetic conditions. So the precise literature-backed statement is this: current primary precedent supports **ATPase assays** and **protein-substrate protease assays**, especially **caseinolytic readouts**, but the full Methods sections still need to be extracted before protocol lock. That is not a flaw in the target; it is simply where the retrievability ceiling landed in this session. [13]

Assay-relevant implications

Assay mode	What is directly supported by retrieved primary sources	Immediate implication
ATPase activity	BB0789 7ZBH construct has ATPase activity; soluble TmFtsH construct in 2CEA was functional in ATPase assays. [14]	Set up ATP hydrolysis as a first orthogonal biochemical readout and a counterscreen for protease-independent effects.
Protease activity	BB0789 7ZBH construct has protease activity; TmFtsH 2CEA used caseinolytic assays. [14]	Start with permissive protein substrates rather than assuming a short fluorogenic peptide will report on the whole machine.
Coupling/ conformation	3KDS links conformational state and linker integrity to function. [10]	Build assays that can distinguish ATPase inhibition, protease inhibition, and ATPase–protease uncoupling.
Oligomer relevance	7ZBH and 2CEA both emphasize hexameric assembly as functionally relevant. [15]	Verify oligomeric state of any recombinant BB0789 construct used for screening; do not assume monomeric constructs are interpretable.

Conservation, homologs, and selectivity

The strongest homology story for BB0789 is not that it is exotic; it is that the **core machine is conserved** and the **best selectivity opportunities probably sit outside the catalytic cores**. The bacterial precedent from **2CEA** shows the same overall logic seen in 7ZBH: a hexameric ATPase/protease assembly, nucleotide-bound ATPase ring, Zn-dependent Asp-zincin protease, and translocation chamber architecture. The apo state in **3KDS** then demonstrates that this machine undergoes nontrivial conformational rearrangements tied to substrate handling. Those papers are the most relevant mechanistic transfer set for BB0789. [16]

On the human side, **2QZ4** is the most directly useful structural warning sign. It solves the **ADP-bound ATPase domain** of human **SPG7/paraplegin**, a mitochondrial m-AAA protease homolog of bacterial FtsH. The abstract makes the family relationship explicit: paraplegin contains an **FtsH-homology protease domain** tandem to an **AAA+ ATPase domain**, and its ATPase conformational cycle is used to deliver substrates to the protease domain. In other words, if you hit the ATPase motor in a generic way, you are not fighting a conceptual off-target risk; you are fighting a documented human homolog class. [17]

Using the deposited sequences from the official RCSB entries for **7ZBH**, **2CEA**, and **2QZ4**, I computed a simple pairwise alignment summary. BB0789 aligns to bacterial *T. maritima* FtsH with about **55.6% identity across aligned residues** and full coverage of the BB0789 construct. BB0789 also aligns to the **human SPG7 ATPase-domain construct** with about **55.6% identity across aligned residues**, covering almost the full SPG7 ATPase construct and about **57%** of the BB0789 construct. That is exactly the pattern you would expect if the ATPase core is conserved while the rest of the bacterial protein supplies the extra discriminatory surfaces. [18]

The motif-level picture agrees with that. In BB0789, the **Walker A/P-loop** motif is around **211–219** and the **pore/threading motif** is around **329–332**; in human SPG7, the ATPase-domain construct preserves a closely related **Walker A** and the same **RPGR-like pore motif** region. Meanwhile the protease active-site ligands in BB0789—**H434, H438, D510**—match the logic seen in the older bacterial structure, where the catalytic zinc is coordinated by **H423, H427, D500**. The unavoidable inference is that **Zn-binding chemistry and ATP-competitive chemistry both start life on conserved ground**. If you want a clean bacterial window, you should look hardest at **Borrelia-specific loop geometry, interdomain linker behavior, oligomer packing, and pore-access topology**. [19]

Selectivity comparison

The table below uses **assistant-computed alignments from RCSB-deposited sequences** plus motif interpretation from the corresponding primary structure entries.

Comparator	Structural basis	Approx. identity to BB0789	What is conserved	What this means for selectivity
Thermotoga maritima FtsH	PDB 2CEA; full soluble bacterial hexamer. [9]	~55.6% across aligned residues	ATPase core, pore motif logic, HExxH protease motif, catalytic Asp-zincin architecture	Good bacterial mechanistic transfer model; supports using 2CEA/3KDS to interpret BB0789 motions.
Human SPG7/ paraplegin ATPase domain	PDB 2QZ4; ADP-bound human mitochondrial homolog ATPase domain. [17]	~55.6% across aligned residues in the ATPase-domain overlap	ATPase P-loop logic and pore/ threading region are strongly conserved	Straight ATP-competitive chemistry is high-risk for mitochondrial off-targets.
Human FtsH-like mitochondrial homolog class	Family relationship described in 2CEA and 2QZ4 abstracts. [20]	Not numerically resolved here for protease domains	Shared AAA+/ FtsH architecture	Protease-site Zn chelation is unlikely to buy selectivity automatically; family-aware counterscreens are mandatory.

Pocket and feature comparison

Because cavity IDs are software- and session-specific, the more durable comparison is by **region** rather than by cavity number.

Region	Zn/ATP proximity	Likely inhibitor bucket	Druggability notes
Catalytic Zn cleft around H434/H438/D510	Directly Zn-proximal	Active-site competitive	Easiest chemistry, worst selectivity risk; structurally real but chemically treacherous. [21]
Nucleotide pocket around Walker A/P-loop	ATP/ADP site	ATP-competitive or nucleotide-state modulators	Mechanistically attractive, but ATPase homology to human m-AAA proteases makes selectivity harder. [22]
Central pore / threading entrance	Functionally adjacent, not directly Zn- or ATP-centered	Pore blockers / substrate-threading inhibitors	Best conceptual selectivity opportunity if pocket volume is real in the hexamer and not an artifact of partial disorder. [23]
ATPase–protease coupling surface	Links nucleotide state to proteolysis	Allosteric uncouplers	Attractive because 3KDS shows conformational coupling is essential. [10]
Intersubunit grooves in the hexamer	Assembly- and motion-dependent	Interface stabilizers or assembly disruptors	Worth ranking high because assembly is functionally required and more species-specific than core active sites. [15]
Flexible unresolved loop belts	Variable; several surround pore/interdomain regions	Cryptic/allosteric pockets	High upside, high modeling uncertainty; should be handled with ensemble methods, not a single rigid receptor. [7]

Inhibitor landscape and druggability

The most honest summary of the directly relevant inhibitor landscape is that it is **immature**. In the core FtsH-family structures examined here—**7ZBH**, **2CEA**, **3KDS**, and **2QZ4**—the resolved small molecules are **ADP**, **Zn**, and associated ions/waters, not development-quality inhibitors. I did not identify a directly retrievable **inhibitor-bound** bacterial FtsH or BB0789 structure in this session. That absence is itself useful information: you are not stepping into a crowded, structurally solved chemotype space. [4]

That said, the structural logic still strongly differentiates candidate inhibition modes. **Zn-site competitive inhibitors** are the obvious first idea because the active site is geometrically real and metal coordination gives potency cheaply. But FtsH was classified as an **Asp-zincin** in the 2CEA paper, and the human mitochondrial m-AAA homolog class is close enough that unselective metal-binding warheads should be considered presumptively dangerous until proven otherwise. Potency is not the hard part there. Specificity is. [24]

ATPase-site inhibitors look superficially more modern, but the selectivity math is not kind. The human SPG7/paraplegin ATPase domain is a direct structural homolog, solved with **ADP** in the nucleotide pocket, and the aligned ATPase region is strongly conserved with BB0789. So if the chemistry idea is merely “bind the P-loop ATPase," that is not a bacterial-selective strategy yet; it is a mitochondrial-toxicity problem that has not happened to you yet. [25]

The more interesting chemotypes are therefore the ones that exploit **state-dependent** or **oligomer-dependent** features: pore blockers, conformation lock-ons, domain-coupling disruptors, or intersubunit interface binders. The apo-vs-ADP contrast in **3KDS vs 2CEA** is the strongest structural argument for that path, because it demonstrates that FtsH function is not just "chemistry at a catalytic center" but coordinated **domain motion**, **pore rearrangement**, and **substrate translocation mechanics**. If you can trap a nonproductive state, you may get selectivity from topology and dynamics instead of from a conserved catalytic atom arrangement. [26]

So the druggability verdict is asymmetric. **The target is druggable. The most obvious pockets are not necessarily the smartest ones.** The Zn site is chemically tractable but selectivity-poor. The ATP pocket is biologically central but human-risky. The pore and coupling interfaces are harder, but they are where a Lyme-directed BB0789 program could plausibly become genuinely distinct rather than just another metalloprotease story with better branding. [27]

Gaps, decisive experiments, and prioritized next steps

The biggest unresolved point is not the structure. It is the **biological proof chain**. The 2024 BB0789 primary paper clearly states that BB0789 is crucial for **mouse infectivity, tick infectivity, and in vitro growth**, but I could not independently retrieve the older direct genetics/infectivity paper during this session. That means the target is promising enough to work on now, but the dossier is not yet as litigation-proof as it should be. The first documentary task is simply to close that citation gap with the original paper in hand. [1]

The second gap is assay granularity. The retrieved primary sources support ATPase measurements, casein-style proteolysis, and the importance of conformational coupling, but not yet a pinned-down BB0789 assay SOP with conditions, substrate design, and expected kinetic behavior. Before a screen starts, the Methods sections behind **7ZBH**, **2CEA**, and **3KDS** should be mined directly for construct boundaries, buffer systems, nucleotide conditions, metals, and readout chemistry. [28]

The third gap is selectivity benchmarking against human homologs. The ATPase warning sign from **SPG7/paraplegin** is already bright enough to shape strategy, but the human **protease-domain** comparison is still under-resolved in this dossier because I did not retrieve a directly comparable human protease-domain structure here. That means ATPase-site risk is already visible; protease-site risk is probably substantial, but still needs sharper structural comparison. [29]

Prioritized actionable next steps

1. **Close the BB0789 biology citation gap.** Retrieve the earlier direct paper underlying the statement that BB0789 is crucial for mouse infectivity, tick infectivity, and in vitro growth, and extract the exact experimental system, genetic perturbation, and phenotype magnitude. This is the single most important literature task because it hardens the target-validation case. [1]

2. **Annotate a standard residue panel in 7ZBH and keep it fixed across all future structure work.** At minimum: **Walker A/P-loop ~211–219**, **threading/pore motif ~329–332**, **Zn-binding protease**

motif H434/H438, and **third catalytic ligand D510**. Those residues should be explicit labels in ChimeraX, PyMOL, docking grids, and mutagenesis plans. [30]

3. **Prepare multiple receptor states, not one.** Use **7ZBH** as the BB0789 anchor, but also interpret pocket behavior through the **ADP-vs-apo** bacterial FtsH comparison (**2CEA vs 3KDS**). The target is conformationally mobile; rigid-receptor docking alone will overfit the wrong geometry. [16]

4. **Stand up two biochemical assays in parallel.** One **ATPase readout** and one **protease readout** built around a permissive unfolded substrate regime. A compound that only kills ATPase activity and a compound that mainly uncouples ATPase from proteolysis are mechanistically different and should not be collapsed into one bin. [28]

5. **Treat Zn chelators as mechanistic tools before treating them as leads.** They can be useful positive controls for "can the protease be inhibited at all," but they should not be granted lead-like status unless selectivity data force that conclusion. [31]

6. **Prioritize three structure-guided screening buckets.**
   First: pore/threading-surface binders.
   Second: ATPase–protease coupling/interface binders.
   Third: intersubunit groove binders in the hexamer.
   Use the Zn site mainly as a benchmark/control bucket, not as the flagship strategy. [32]

7. **Build human-homolog counterscreens early.** At minimum, any ATPase-directed series should be stress-tested conceptually against **SPG7/paraplegin-like** ATPase conservation from day one, not after chemistry has already emotionally committed to the scaffold. [17]

Recommended reading

The shortest high-value reading stack is:

- **Brangulis et al. 2024** — the BB0789 paper behind **7ZBH**; this is the direct target anchor. [1]
- **Bieniossek et al. 2006** — **2CEA**, the foundational bacterial FtsH architecture paper. [9]
- **Bieniossek et al. 2009** — **3KDS**, the apo-state conformational paper. [10]
- **Karlberg et al. 2009** — **2QZ4**, the human SPG7/paraplegin ATPase-domain comparator. [17]

If you want the one-sentence strategic takeaway after all of that: **BB0789 is worth pursuing, but only if the campaign is disciplined enough to chase bacterial topology and conformational coupling instead of falling in love with generic metal binding or generic ATP mimicry.** [27]

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[1] [2] [4] [5] [7] [8] [14] [15] [18] [21] [22] [27] [28] [30] https://www.rcsb.org/structure/7ZBH
https://www.rcsb.org/structure/7ZBH

[3] [11] [17] [25] [29] https://www.rcsb.org/structure/2QZ4
https://www.rcsb.org/structure/2QZ4

[6] [9] [12] [13] [16] [19] [20] [24] [31] https://www.rcsb.org/structure/2CEA
https://www.rcsb.org/structure/2CEA

[10] [23] [26] [32] https://www.rcsb.org/structure/3KDS
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